

Matrix Row Operations

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1 Introduction

Row operations are a collection of operations that can be performed on a matrix without changing the information the matrix holds. As we know matrices obey linearity and as such we can perform operations on them that respect linearity.

2 Definitions and Theorems

Elementary Row Operations

Let $a, c \in \mathbb{R}$ and $c \neq 0$.

(Replacement) $R_i \rightarrow R_i + aR_j$

(Interchange) $R_i \leftrightarrow R_j$

(Scaling) $R_i \rightarrow cR_i$

Lets break down what each of these are.

Replacement

Replacement is taking a given row of a matrix, row i (R_i) and replacing R_i by the sum of row i and a scalar multiple of row j . That is to say $R_i \rightarrow R_i + aR_j$.

Interchange

Interchange is switch two rows with each other. The i th row becomes the j th row and the j th row becomes the i th row. $R_i \leftrightarrow R_j$

Scaling

Scaling takes the i th row and multiplies by some nonzero scalar. written as $R_i \rightarrow cR_i$.

3 Examples

When we perform row operations we respect the information in the matrix. This is because all row operations respect linear operations, but not in math speak these are all things we can do when solving linear equations. Replacement is like adding two equations. Interchange is like reordering two linear equations. And Scaling is multiplying an equation by a constant.

First we will look at an example of each individual operation. Then we will solve the same system we saw in linear combinations this time using row operations.

Consider the 3x3 matrix

$$\begin{bmatrix} 42 & 87 & 19 \\ 63 & 91 & 35 \\ 74 & 58 & 26 \end{bmatrix}$$

Replacement $R_i \rightarrow R_i + aR_j$:

we will take $R_2 \rightarrow R_2 + 5R_1$

$$\begin{bmatrix} 42 & 87 & 19 \\ 63 & 91 & 35 \\ 74 & 58 & 26 \end{bmatrix} \cong \begin{bmatrix} 42 & 87 & 19 \\ 273 & 526 & 130 \\ 74 & 58 & 26 \end{bmatrix}$$

Interchange $R_i \leftrightarrow R_j$:

we will take $R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 42 & 87 & 19 \\ 63 & 91 & 35 \\ 74 & 58 & 26 \end{bmatrix} \cong \begin{bmatrix} 74 & 58 & 26 \\ 63 & 91 & 35 \\ 42 & 87 & 19 \end{bmatrix}$$

Scaling $R_i \rightarrow cR_i$:

we will take $R_3 \rightarrow \frac{1}{2}R_3$

$$\begin{bmatrix} 42 & 87 & 19 \\ 63 & 91 & 35 \\ 74 & 58 & 26 \end{bmatrix} \cong \begin{bmatrix} 42 & 87 & 19 \\ 63 & 91 & 35 \\ 37 & 29 & 13 \end{bmatrix}$$

Now that we have seen each row operation n lets apply them to solve a problem! Recall the system, or don't if you didn't read part 1 of the linear algebra overview.

$$42x + 67y + 73z = \ln(\pi)$$

$$\sin(4)x - y + 0z = 0$$

$$-x + 2y + 24z = 6$$

Turning this into an augmented matrix we find

$$\left[\begin{array}{ccc|c} 42 & 67 & 73 & \ln(\pi) \\ \sin(4) & -1 & 0 & 0 \\ -1 & 2 & 24 & 6 \end{array} \right]$$

There are many options of how to start, but our goal is to turn this augmented matrix into reduced row echelon form. Then we can clearly see what x, y, z are.

Personally I see row 3 has a one in the first coulomb so lest start with an interchange.

$$R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 42 & 67 & 73 & \ln(\pi) \\ \sin(4) & -1 & 0 & 0 \\ -1 & 2 & 24 & 6 \end{array} \right] \cong \left[\begin{array}{ccc|c} -1 & 2 & 24 & 6 \\ \sin(4) & -1 & 0 & 0 \\ 42 & 67 & 73 & \ln(\pi) \end{array} \right]$$

Now lets use scaling on row 1.

$$R_1 \rightarrow -R_1$$

$$\left[\begin{array}{ccc|c} -1 & 2 & 24 & 6 \\ \sin(4) & -1 & 0 & 0 \\ 42 & 67 & 73 & \ln(\pi) \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ \sin(4) & -1 & 0 & 0 \\ 42 & 67 & 73 & \ln(\pi) \end{array} \right]$$

Lets now perform replacement on row 3 to get aero in the first coulomb.

$$R_3 \rightarrow R_3 - 42R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ \sin(4) & -1 & 0 & 0 \\ 42 & 67 & 73 & \ln(\pi) \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ \sin(4) & -1 & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right]$$

Lets scale row 2.

$$R_2 \rightarrow \frac{1}{\sin(4)} R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ \sin(4) & -1 & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 1 & \frac{-1}{\sin(4)} & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right]$$

But we have the trig identity $\frac{1}{\sin(x)} = \csc(x)$ so now instead of a congruence we have an equivalence. But we can

leverage a fact that if two things are equivalent they are congruent.

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 1 & \frac{-1}{\sin(4)} & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 1 & \csc(4) & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right]$$

Lets do row replacement on row 2.

$$R_2 \rightarrow R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 1 & \csc(4) & 0 & 0 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & \csc(4) + 2 & 24 & 6 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right]$$

lets scale row 2

$$R_2 \rightarrow \frac{1}{\csc(4)+2} R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & \csc(4) + 2 & 24 & 6 \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right]$$

Doing row replacement on row 3.

$$R_3 \rightarrow R_3 - 151R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 151 & 1061 & \ln(\pi) + 252 \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & \frac{1061(\csc(4)+2)-3624}{\csc(4)+2} & \frac{(\ln(\pi)+252)(\csc(4)+2)-906}{\csc(4)+2} \end{array} \right]$$

simplifying we have another equivalence relation we will downgrade to a congruence relation

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & \frac{1061(\csc(4)+2)-3624}{\csc(4)+2} & \frac{(\ln(\pi)+252)(\csc(4)+2)-906}{\csc(4)+2} \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & \frac{1061 \csc(4)-1502}{\csc(4)+2} & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{\csc(4)+2} \end{array} \right]$$

lets now scale row 3.

$$R_3 \rightarrow \frac{\csc(4)+2}{1061 \csc(4)-1502} R_3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & \frac{1061 \csc(4)-1502}{\csc(4)+2} & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{\csc(4)+2} \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right]$$

Lets do row replacement on row 1.

$$R_1 \rightarrow R_1 + 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -24 & -6 \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & 0 & -24 + \frac{48}{\csc(4)+2} & -6 + \frac{12}{\csc(4)+2} \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right]$$

simplifying

$$\left[\begin{array}{ccc|c} 1 & 0 & -24 + \frac{48}{\csc(4)+2} & -6 + \frac{12}{\csc(4)+2} \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & 0 & \frac{-24 \csc(4)}{\csc(4)+2} & \frac{-6 \csc(4)}{\csc(4)+2} \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right]$$

Now for the big two finishing moves! Lets do row replacement on row 1.

$$R_1 \rightarrow R_1 + \frac{-24 \csc(4)}{\csc(4)+2} R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{-24 \csc(4)}{\csc(4)+2} & \frac{-6 \csc(4)}{\csc(4)+2} \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right] \cong \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{\csc(4)(-6-24[\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402])}{(1061 \csc(4)-1502)(\csc(4)+2)} \\ 0 & 1 & \frac{24}{\csc(4)+2} & \frac{6}{\csc(4)+2} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right]$$

Ending with row replacement for row 2 we have

$$R_2 \rightarrow R_2 - \frac{24}{\csc(4)+2} R_3$$

$$\cong \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{\csc(4)(-6-24[\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402])}{(1061 \csc(4)-1502)(\csc(4)+2)} \\ 0 & 1 & 0 & \frac{6}{\csc(4)+2} - \frac{24}{\csc(4)+2} \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \\ 0 & 0 & 1 & \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502} \end{array} \right]$$

so our end results are

$$x = \frac{\csc(4)(-6-24[\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402])}{(1061 \csc(4)-1502)(\csc(4)+2)}$$

$$y = \frac{6}{\csc(4)+2} - \frac{24}{\csc(4)+2} \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502}$$

$$z = \frac{\ln(\pi) \csc(4)+252 \csc(4)+2 \ln(\pi)-402}{1061 \csc(4)-1502}$$

For your practice I leave it to you to put these in the form found in part 1 of the linear algebra overview.